Complex Disordered Systems

Polymers and solvents

Francesco Turci

Today

- Good, poor and theta solvents
- Concentrated polymer solutions

Polymers in solvents

- Polymers typically are dispersed in a medium (**solvent**).
- Solvent-polymer attractions (the **affinity**) influences the polymer conformations:
  - **good solvents** $\longrightarrow$ high affinity $\longrightarrow$ the polymer swells (expands)
  - **poor solvents** $\longrightarrow$ low affinity $\longrightarrow$ the polymer shrinks

  The affinity depends on the details of the molecular interactions but can be *coarse-grained*.

  We can construct a simple model with **effective interaction energies**
- ε_{ss} for the solvent-solvent
- $\varepsilon_{\mathrm{pp}}$ for monomer-monomer
- ε_{sp} for solvent-monomer interactions.

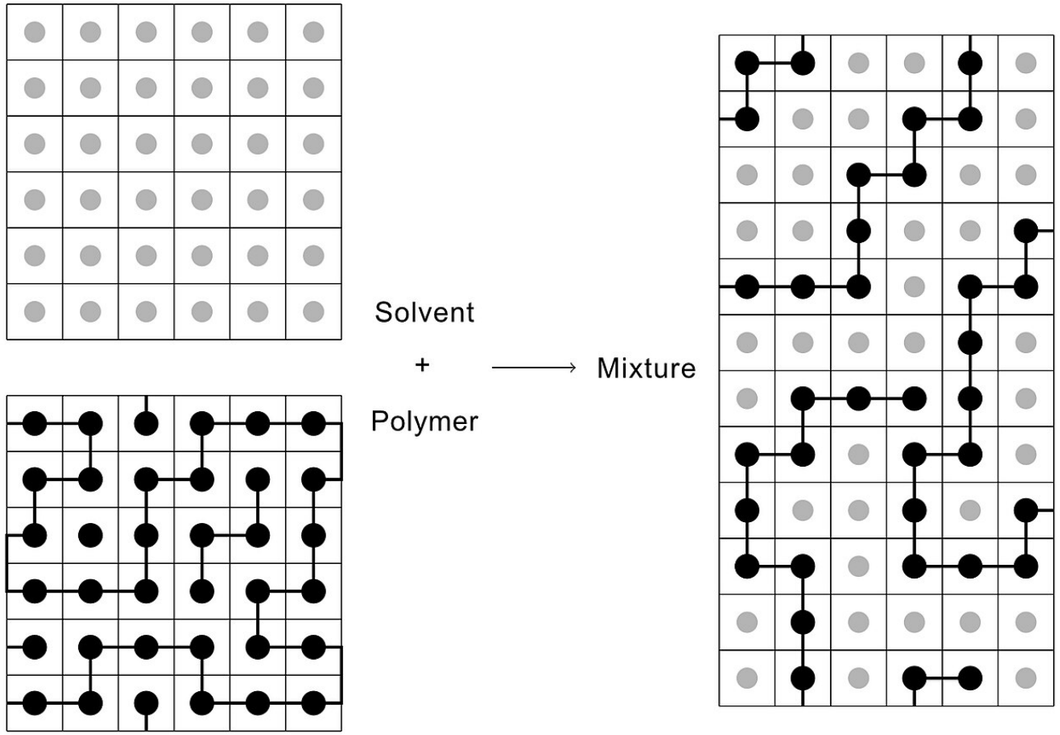
Energy of mixing

On-lattice model:

- each site has z nearest neighbours
- N_s solvent molecules
- N_p monomers
- $\mathrm{N_{sp}}$ solvent-monomer contacts.

Effective interactions:

- ε_{ss} for the solvent-solvent
- $\varepsilon_{\mathrm{pp}}$ for monomer-monomer
- ε_{sp} for solvent-monomer interactions.



Then the energy of mixing ΔU_{mix} is obtained as the difference between the total energy U of the mixed system and the sum of the energies of the **pure solvent** and **pure polymer**, $U_S + U_p$.

$$\Delta U_{\mathrm{mix}} = U - (U_S + U_p)$$

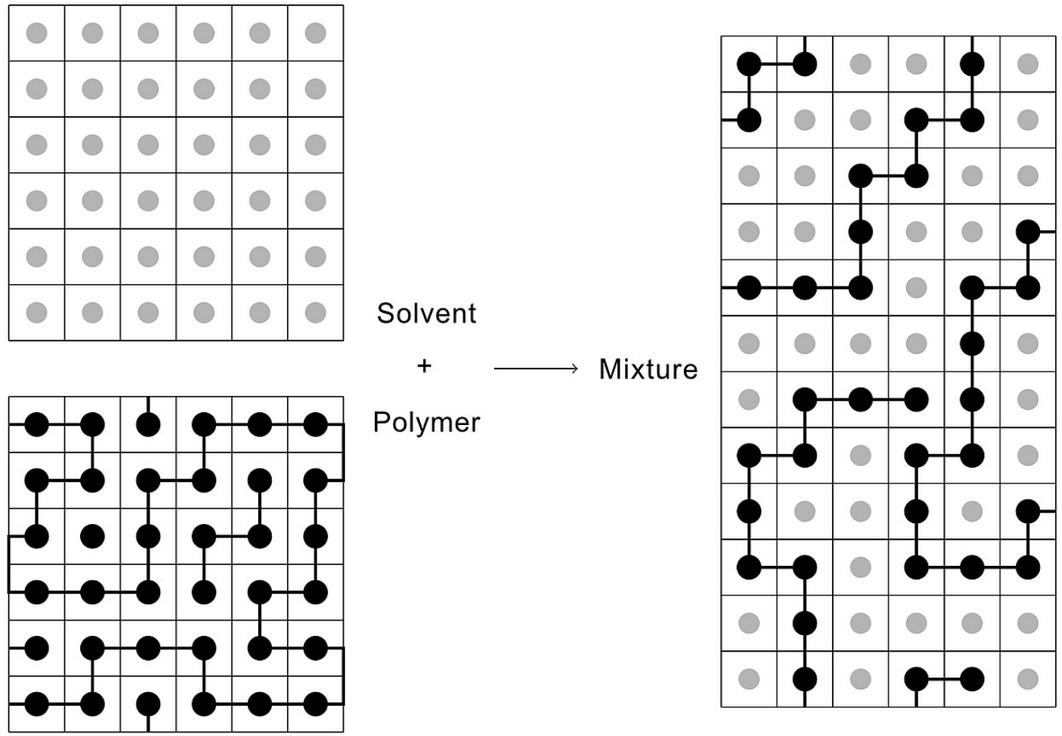
Energy of mixing

- Energy of pure solvent

$$U_s = \frac{zN_s\varepsilon_{ss}}{2}$$

- Energy of pure polymer

$$U_p = \frac{zN_p\varepsilon_{pp}}{2}$$



Energy of solution $U = N_{sp}\varepsilon_{sp} + \frac{(zN_s - N_{sp})\,\varepsilon_{ss}}{2} + \frac{(zN_p - N_{sp})\,\varepsilon_{pp}}{2}$

We obtain that the **energy of mixing** $\Delta U_{\text{mix}} = N_{sp}\left[\varepsilon_{sp} - \frac{1}{2}\left(\varepsilon_{ss} + \varepsilon_{pp}\right)\right] \equiv N_{sp}\,\Delta\varepsilon$

which can **change sign** depending on our choices for the effective interaction energies.

Excluded volume vs mixing energy

- Each lattice site holds **one** molecule: the **excluded volume**, an **entropic** cost $\sim k_B T$ per monomer-monomer contact.
- For $\Delta\varepsilon > 0$ the same contact is **energetically favoured**, by $\sim z\,\Delta\varepsilon$.
- The net cost per contact sets the **excluded-volume parameter** v of the Flory free energy:

$$v \simeq b^3 \left(1 - \frac{\Theta}{T}\right), \qquad \Theta \simeq \frac{z\,\Delta\varepsilon}{k_B}$$

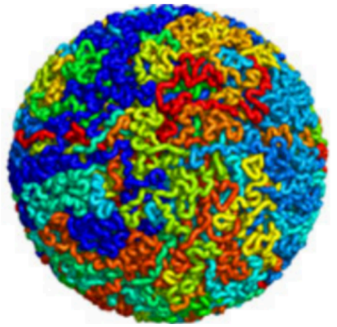
Solvent quality is the **sign of** v.

Good solvents, poor solvents and theta solvents

$\frac{F}{k_B T} \simeq \frac{3R^2}{2Nb^2} + v\,\frac{N^2}{R^3} + w\,\frac{N^3}{R^6}$: keep the terms that matter.

Solvent	v	Balance	Size
Good, $T > \Theta$	$v > 0$	elastic vs two-body	$R \sim N^{3/5}$
Theta, $T = \Theta$	$v = 0$	elastic vs three-body	$R \sim N^{1/2}$
Poor, $T < \Theta$	$v < 0$	two-body vs three-body	$R \sim N^{1/3}$

Collapsed chains: hydrophobic polypeptides, polystyrene in water, chromatin in the cell nucleus.



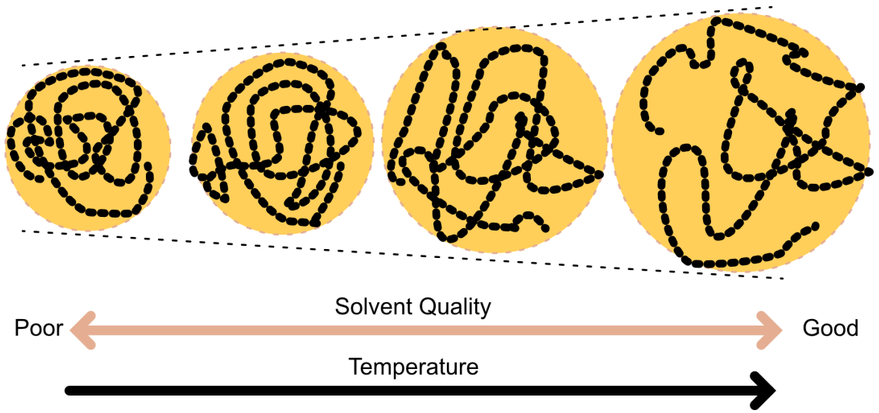
Chromatin globule model

- Tuning solvent properties (e.g. the temperature) leads to **conformational changes**

Temperature dependence and theta (Θ) solvents

The various affinities have temperature dependencies. Typically:

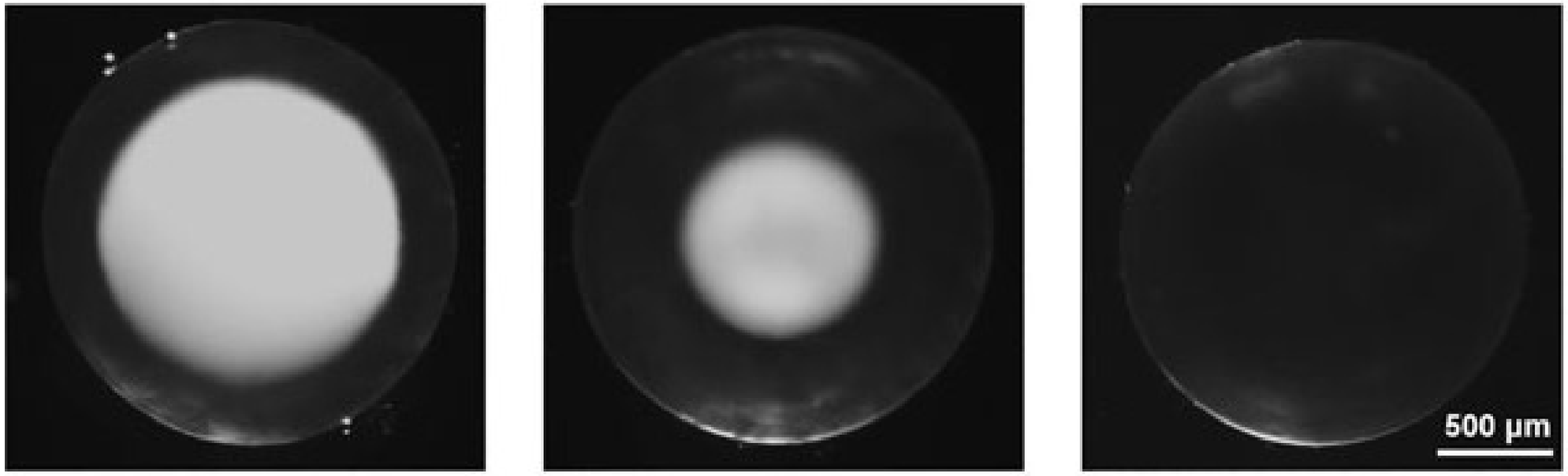
- At high T, the coil expand and the solvent is *good*
- At low T the coil collapses and phase separation is observed (polymer-rich from polymer-poor phase)
- There is an intermediate temperature where the excluded volume and attractive interaction compensate each other ($v = 0$). The three-body term then gives $R^8 \sim N^4$: in $d = 3$ the polymer behaves like an ideal chain (freely jointed chain), $R \sim N^{1/2}$.
  - The temperature is conventionally named Θ **temperature** (or *Flory* temperature, after physicist Paul Flory)
  - The Θ temperature increases with the effective monomer-monomer attractions (we need higher temperature to reach ideal behaviour)



Swelling behaviour of polymers in solvent.

Examples

- Example: **cold cataract**. The eye lens is a concentrated solution of proteins (crystallins) in water.
  - At body temperature water is a good solvent: the lens is transparent.
  - On cooling, water becomes a poorer solvent: the mixture separates into protein-rich and protein-poor regions, which scatter light, and the lens turns opaque.
  - Warming back to $37\,^{\circ}\mathrm{C}$ clears the lens within minutes: the change is reversible.



Mouse lenses at $4\,^{\circ}\mathrm{C}$, $25\,^{\circ}\mathrm{C}$ and $37\,^{\circ}\mathrm{C}$ (left to right). Images from Li *et al.*, *Front. Chem.* **9**, 767696 (2021), CC BY. Phase separation in cold cataract: Tanaka, Ishimoto & Chylack, *Science* **197**, 1010 (1977).

Concentrated polymer solutions

- We mostly dealt with **single polymers** and their properties up to now.
- Many polymers together make a **polymer solution**

As we vary the **concentration** the behavior qualitatively changes.

We will increase the concentration in steps to reach **bulk polymers**

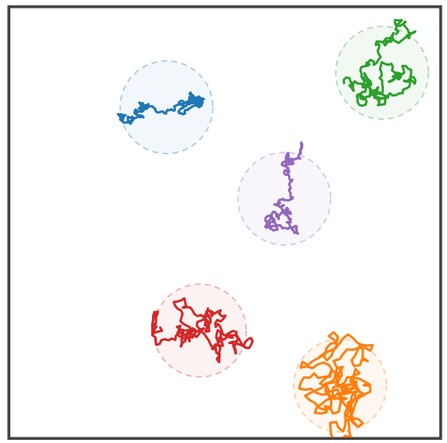
Dilute regime

- The polymer coils are well-separated on average.
- The system is dilute if the polymer concentration c is such that

$$\left(\frac{c}{M}N_A\right)\left(\frac{4\pi}{3}R_g^3\right) \ll 1$$

where N_A is Avogadro's number, M is the molar mass of a single chain and R_g is the radius of gyration of the chain.

Essentially, the polymers **do not overlap**.



dilute c ≪ c*

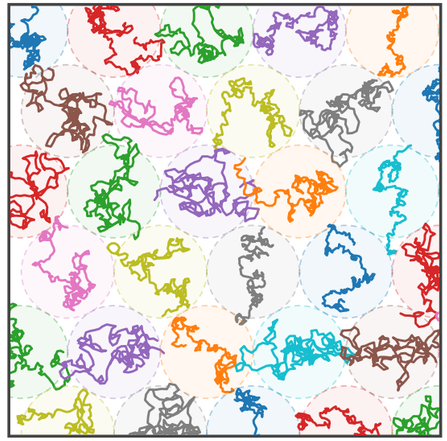
Overlap concentration $c*$

Overlap occurs when the volume fraction of coils reaches unity and thus

$$\frac{c^*}{M} N_A \frac{4\pi}{3} R_g^3 \sim 1 \quad \therefore c^* = \frac{3M}{4\pi N_A R_g^3}$$

Phenomenologically one has $Rg = \langle R_g^2 \rangle^{1/2} = BM^\nu$ with an empirical exponent called the **Flory scaling exponent**. Hence

$$c^* = \frac{3}{4\pi N_A B^3} M^{1-3v}$$



overlap c ≈ c*

Overlap concentration $c*$

$$c^* = \frac{3}{4\pi N_A B^3} M^{1-3v}$$

Example of polystyrene in a good solvent:

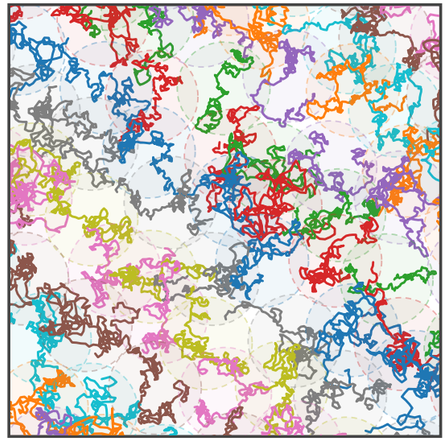
Parameter	Value	Unit
Molar mass (M)	10^6	g mol^{-1}
Flory exponent (v)	0.6	
Prefactor (B)	0.028	nm (g mol^{-1})$^{-0.6}$
Overlap concentration (c*)	0.29	kg m^{-3} (or 0.29 mg/ml)
Polystyrene density (ρ)	1050	kg m^{-3}
Volume fraction of monomers	0.28×10^{-3}	

For large polymers, c^* can be very small.

Semi-dilute

- The concentration is larger than the overlap concentration c^*, but still much smaller than the bulk density.
- The coils interpenetrate and **entanglement** begins
- Change in the dynamics (slowing down)
- Yet, the solution is still mostly solvent.

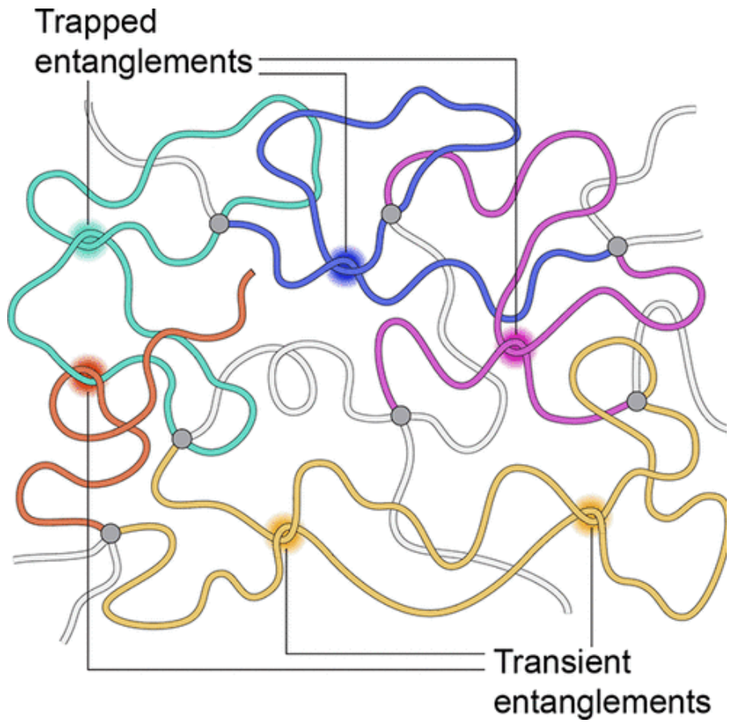
Start of the formation of a **polymer network** with transient **crowded regions** *within* the solvent.



semi-dilute c > c*

Polymer entanglement

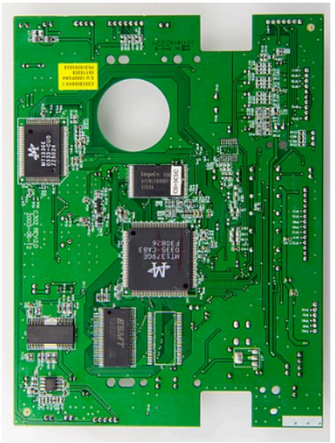
- Overlapping polymer chains become topologically **intertwined**
- Creates **physical knots** that restrict chain motion
  - changes flow behaviour (*rheology*)
  - changes material properties (e.g. elasticity, viscosity)
- Introduces **topological constraints** (transient/permanent)



Concentrated: bulk polymers

- Polymer concentration approaches **bulk density**: **highly interpenetrated coils**, entanglements restrict segment motion, **solid-like** behaviour.
- **Cross-linking** forms **covalent bonds between separate polymer chains**, creating a **three-dimensional network**. Three main classes of bulk polymers, characterised by their degree of cross-linking:
  - **Elastomers** (rubbers): **low cross-linking** → flexible and elastic (natural rubber, silicone)
  - **Thermosets**: **high cross-linking** → rigid, hard, brittle network (epoxy resins)
  - **Thermoplastics**: **no cross-linking** → most everyday plastics (PE, PET, PS); behaviour upon cooling shows similarities to colloids



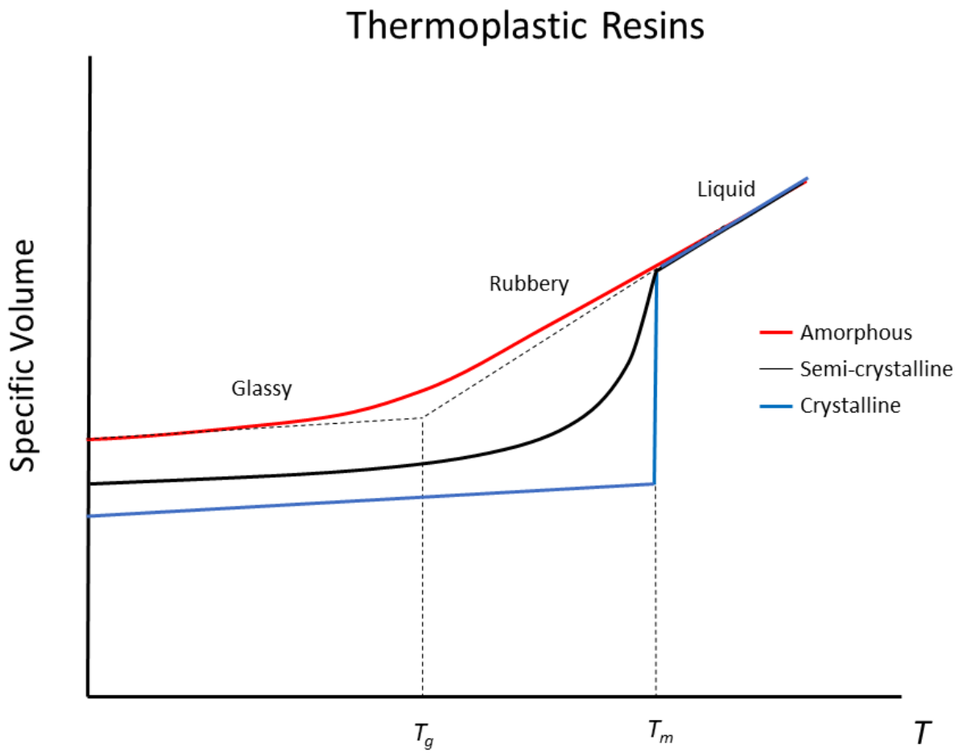




Natural rubber, epoxy substrate for circuits, PET used in plastic bottles

Thermoplastics behaviour upon cooling

- At high temperature the free energy is dominated by **entropy**.
  - random assembly of intertwined flexible coils
- Upon cooling, potential energy dominates and chain rotation becomes restricted, favoring more extended configurations.
- Below melting temperature T_m, **crystals form** (lowest free energy state), but require slow cooling and significant molecular ordering.
- Rapid cooling below glass transition temperature T_g ($< T_m$) **prevents crystallization**, forming an **amorphous glassy state** instead—a metastable, long-lived solid.
- Solid thermoplastics typically contain a **mixture of crystalline and amorphous regions**.



Summary: polymers

Topic	Key result
Freely-jointed chain	$\langle R^2 \rangle = N b_0^2$, Gaussian $P(\mathbf{R})$: a chain is a random walk
Radius of gyration	$\langle R_g^2 \rangle = \langle R^2 \rangle / 6$
Freely-rotating chain	$\langle R^2 \rangle = C N b_0^2$, $C = \frac{1+\cos\theta}{1-\cos\theta}$: stiffness only changes the prefactor
Kuhn length	$b = C b_0$, $N' = N/C$: chemistry is hidden in b
Excluded volume	$R \sim N^{3/5}$ (self-avoiding walk)
Solvent quality	$v \simeq b^3(1 - \Theta/T)$: excluded volume vs mixing energy; in $d = 3$, good $\nu \approx 3/5$; Θ $\nu = 1/2$; poor: collapsed globule $\nu = 1/3$
Overlap concentration	$c^* = \frac{3M}{4\pi N_A R_g^3} \propto M^{1-3\nu}$, very small for long chains
Semi-dilute	$c > c^*$: interpenetration, entanglement, slow flow
Bulk polymers	elastomers, thermosets, thermoplastics; on cooling crystal (slow) or glass (fast)

Full summary with links in the notes